LESSON 1 · FIND THE AREA OF A PARALLELOGRAM

Two Sides, Two Heights, Same Area

E L A + M A T H · W O R D S T U D Y

Mr. H Math · Chandler Park Academy · Day 9

CCSS
6.G.A.1
3.MD.C.7d
SMP 2,4,5,6

WRITE YOUR NAME — one letter per box, straight and neat

Date: _____ **Score: ___ / 10**

Yesterday you proved that a leaning parallelogram is really a rectangle in disguise. Today's i-Ready **Refine** problems and Lesson 1 Quiz use that same idea in trickier ways — a mixed-number base, a classmate's mistake to catch, and a parallelogram with two different sides that are BOTH honest bases. Every blank you fill in unlocks a word from today's vocabulary — read closely and choose your words carefully!

Section A · Guess the Word

Fill in the blanks to find the word that means "meeting to form a right angle."

[ ] E R [ ] E N D I C U L A R

The SAME missing letter goes in both dashed boxes. Write it here: _____

The height of a parallelogram must always be drawn _____ to the base — meeting it at a right angle — never along the slanted side.

A parallelogram's height and base always meet to form a right angle. That right angle is marked with a small square in every diagram.

Section B · Meet the Vocabulary

Use the word bank to complete each sentence.

parallelogram · **base** · **height** · **perpendicular**

1. A quadrilateral with opposite sides parallel and equal in length is called a _____.
   Example: a leaning rectangle, a tipped-over box shape.
2. The side of a parallelogram that you choose to measure along, and then multiply by its matching height, is called the _____.
   Example: the 9 cm side the shape is "standing" on.
3. The straight, perpendicular distance from the base to the opposite side — never the slanted side — is called the _____.
   Example: marked with a small right-angle square, not measured along the slant.
4. A parallelogram's base and its matching height always meet at a right angle — we say the height is _____ to the base.
   Example: this is the same word you unlocked in Section A.

Section C · Try It: One Parallelogram, Two Correct Heights

Study the worked example, then solve your own. A parallelogram has two pairs of parallel sides — each pair has its own honest base-and-height pair, and both give the exact same area.

WORKED EXAMPLE

Area = 24 sq units

One pair of sides = 6 units · other pair = 4 units

If base = 6: height = 24 ÷ 6 = 4 units

If base = 4: height = 24 ÷ 4 = 6 units

↑ same parallelogram ↑ same area, either way

YOUR TURN

A parallelogram has an area of 30 sq units. One pair of sides is 5 units long. The other pair is 6 units long. Find BOTH possible heights, then draw the parallelogram in the grid and label the base and height you used.

If base = 5 units: height = 30 ÷ 5 = _____ units

If base = 6 units: height = 30 ÷ 6 = _____ units

REFINE · SUMMARY — A parallelogram doesn't have just one base and one height: it has two, one for each pair of parallel sides — and both multiply to the exact same area. Watch, too, for a slanted side or a diagonal printed right in the problem — it's real, but it's never the number you multiply. Next, the Area unit turns to **triangles** — cut a parallelogram in half along a diagonal, and each half is exactly one triangle, with exactly half the area.

☆ Two sides, two heights, one same area! ☆

E L A + M A T H